



# Chapter 1: Introduction

|                     |                      |
|---------------------|----------------------|
| 🕒 Created by        | 🌐 Aroosh Ahmad       |
| 🕒 Created time      | May 18, 2026 7:47 AM |
| 🕒 Last edited by    | 🌐 Aroosh Ahmad       |
| 🕒 Last updated time | May 18, 2026 7:48 AM |

## 1.1 Background

Large Language Models (LLMs) have become a central area of research in Natural Language Processing due to their ability to perform a wide range of language understanding and generation tasks. Pre-trained language models are typically trained on large-scale text corpora and learn general linguistic, semantic, and contextual patterns from data. However, pre-training alone does not necessarily make a model capable of following human instructions effectively. A model may generate fluent text, but still fail to respond according to the user's intent, task requirement, or expected output format.

Instruction fine-tuning has emerged as an important method for improving the usability of language models. Instead of training a model only to predict the next token or reconstruct masked text, instruction fine-tuning adapts the model using instruction-response pairs. This allows the model to learn how to respond to natural language commands, questions, and task descriptions. Prior work such as InstructGPT demonstrated that models fine-tuned to follow instructions can become more aligned with user intent, while FLAN-style instruction tuning showed that fine-tuning across diverse tasks can improve generalization to unseen tasks.

Despite these advances, instruction fine-tuning remains sensitive to several design choices, including dataset quality, task diversity, model size, optimization configuration, and training strategy. In most supervised fine-tuning pipelines, training examples are divided into mini-batches using random sampling. This is

usually treated as a standard implementation detail rather than a research variable. However, the composition and ordering of mini-batches may influence the gradients seen by the model during training, and therefore may affect convergence behavior, generalization, and final response quality.

This thesis investigates whether mini-batch construction can play a meaningful role in instruction fine-tuning. In particular, it studies semantic batching, where semantically similar instructions are grouped together, and curriculum-based batching, where training examples are ordered according to a defined schedule.

---

## 1.2 Instruction Fine-Tuning of Large Language Models

Instruction fine-tuning is the process of adapting a pre-trained language model using datasets that contain instructions and corresponding responses. The goal is to make the model better at understanding task descriptions and producing outputs that match user expectations. This approach is especially important for general-purpose language models because users interact with such systems through natural language instructions rather than fixed task labels.

Datasets used for instruction fine-tuning often contain diverse categories of tasks such as question answering, summarization, classification, brainstorming, information extraction, and open-ended generation. For example, the Databricks Dolly 15k dataset consists of instruction-following records across multiple behavioral categories inspired by instruction-following research.

In this study, subsets of the Dolly dataset are used to evaluate the effect of different batching strategies. The focus is not on introducing a new model architecture, but on analyzing whether the way training samples are grouped and ordered can influence instruction fine-tuning behavior.

---

## 1.3 Parameter-Efficient Fine-Tuning

Fine-tuning large language models by updating all model parameters can be computationally expensive. Parameter-efficient fine-tuning methods address this problem by training only a small number of additional or selected parameters while keeping most of the pre-trained model frozen. One widely used approach is Low-Rank Adaptation (LoRA), which inserts trainable low-rank matrices into the model and significantly reduces the number of trainable parameters.

This thesis uses LoRA-based fine-tuning with Flan-T5-small. Flan-T5 is suitable for instruction fine-tuning research because it belongs to the family of instruction-tuned encoder-decoder models and has publicly available checkpoints. The use of LoRA makes repeated experiments feasible under limited computational resources while still allowing meaningful comparison across batching strategies.

---

## 1.4 Mini-Batch Construction in Fine-Tuning

In neural network training, mini-batches are used to estimate gradients and update model parameters. In most training pipelines, mini-batches are constructed randomly. Random batching has several advantages: it provides diversity within each update, reduces ordering bias, and often acts as a form of implicit regularization.

However, random batching is not the only possible strategy. Mini-batches can also be constructed according to similarity, difficulty, task type, or curriculum schedule. In the context of instruction fine-tuning, this raises an important question: should semantically similar instructions be trained together, or is random mixing more beneficial?

Semantic batching is based on the intuition that examples with similar meaning may produce more coherent gradient updates. If instructions within a batch are semantically related, the model may receive a more consistent learning signal. On the other hand, instruction-tuning datasets are highly heterogeneous, and excessive grouping may reduce gradient diversity. Therefore, the effect of semantic batching is not obvious and requires empirical investigation.

---

## 1.5 Research Motivation

The motivation for this research comes from a gap between common training practice and the potential importance of data organization. While considerable work has been done on improving model architecture, scaling datasets, filtering data, and designing instruction templates, the role of mini-batch construction is often overlooked.

In standard supervised fine-tuning, the assumption is that if the dataset and optimization settings are fixed, random batching is sufficient. This thesis

challenges that assumption by treating batch construction as an experimental variable.

The central motivation is to understand whether a simple change in the training process—without modifying the model architecture or increasing dataset size—can influence learning. If semantic or curriculum-based batching improves performance, it may offer a low-cost method for improving instruction fine-tuning. If it does not improve performance, that result is also valuable, because it helps clarify the limits of batch-level structuring and reinforces the strength of random batching as a baseline.

---

## 1.6 Problem Statement

Instruction fine-tuning is commonly performed using randomly constructed mini-batches, but the effect of mini-batch composition and ordering on training dynamics is not well understood. In particular, it remains unclear whether semantically coherent batches or curriculum-based ordering can improve optimization behavior, generalization, or generation quality compared to standard random batching.

This thesis addresses the following problem:

**To investigate whether semantic mini-batch construction and curriculum-based batch scheduling can improve instruction fine-tuning of large language models compared to random batching.**

---

## 1.7 Research Gap

Existing work in instruction fine-tuning has largely focused on data quality, task diversity, model scale, alignment, and parameter-efficient adaptation. Curriculum learning has also been studied in machine learning, but the specific role of mini-batch construction in instruction fine-tuning remains less explored.

The gap addressed in this study is the lack of systematic empirical analysis of how batch composition and batch ordering affect instruction fine-tuning. In particular, this thesis examines whether semantically grouped mini-batches and curriculum schedules provide measurable benefits over random batching across different dataset scales and random seeds.

---

## 1.8 Research Objectives

The main objective of this thesis is to empirically evaluate the effect of different batch scheduling strategies on instruction fine-tuning.

The specific objectives are:

1. To implement and evaluate random batching as a baseline for instruction fine-tuning.
  2. To construct semantically grouped mini-batches using sentence embeddings and nearest-neighbor retrieval.
  3. To compare static batching strategies, including random, grouped, and mixed batching.
  4. To investigate curriculum-based ordering strategies such as Random → Grouped and Grouped → Random.
  5. To evaluate difficulty-based curriculum strategies using instruction length as a difficulty proxy.
  6. To analyze the effect of batching strategies across multiple dataset sizes and random seeds.
  7. To compare methods using both optimization metrics and generation quality metrics.
- 

## 1.9 Research Questions

This thesis is guided by the following research questions:

**RQ1:** Does semantic grouping of instructions improve instruction fine-tuning performance compared to random batching?

**RQ2:** Does the ordering of batching strategies during training affect learning dynamics?

**RQ3:** Is random batching a sufficiently strong baseline for instruction fine-tuning?

**RQ4:** Do curriculum-based strategies provide more consistent benefits than static semantic grouping?

**RQ5:** Can length-based difficulty ordering improve evaluation loss and generation quality?

---

## 1.10 Research Hypotheses

The study is based on the following hypotheses:

**H1:** Semantically grouped mini-batches may produce more coherent gradient updates and improve instruction fine-tuning performance.

**H2:** Curriculum ordering may influence training dynamics more effectively than static batch composition.

**H3:** A Random → Grouped schedule may support an exploration–refinement pattern, where early random batching encourages broad task learning and later grouped batching supports focused refinement.

**H4:** Length-based curriculum learning may provide a stronger signal than semantic similarity alone because instruction length can act as a proxy for task difficulty.

These hypotheses are tested empirically using controlled experiments across multiple dataset sizes and random seeds.

---

## 1.11 Scope of the Study

This thesis focuses on supervised instruction fine-tuning using Flan-T5-small with LoRA. The experiments are conducted on Dolly dataset subsets of approximately 1k, 3k, and 5k examples. The study compares random batching, semantic grouping, mixed batching, two-phase curriculum strategies, and length-based curriculum schedules.

The scope is limited to controlled fine-tuning experiments and automated evaluation metrics. The study does not attempt to train a new foundation model, introduce a new architecture, or perform large-scale human evaluation. Instead, it focuses specifically on whether batch construction and data ordering affect fine-tuning behavior under practical resource constraints.

---

## 1.12 Significance of the Study

This research is significant for three main reasons.

First, it examines a training design choice that is often treated as a minor implementation detail. If batch construction affects learning, then fine-tuning pipelines can be improved without requiring larger models or larger datasets.

Second, the study provides a careful empirical evaluation of semantic batching. Even if semantic batching does not consistently improve performance, the result is still meaningful because it challenges an intuitive assumption and clarifies the limits of semantic grouping in instruction fine-tuning.

Third, the study explores curriculum-based scheduling as a more promising direction. By comparing semantic grouping with length-based curriculum learning, the thesis contributes insight into whether difficulty-aware ordering may be more useful than semantic similarity alone.

Overall, the study contributes to a better understanding of training dynamics in instruction fine-tuning and provides evidence that can guide future research on curriculum design, data ordering, and efficient fine-tuning strategies.

---

## 1.13 Thesis Organization

The rest of the thesis is organized as follows.

**Chapter 2** presents the literature review, covering large language models, instruction fine-tuning, LoRA, curriculum learning, semantic similarity, and evaluation metrics.

**Chapter 3** describes the methodology, including the research design, batching strategies, semantic grouping procedure, curriculum schedules, and evaluation protocol.

**Chapter 4** presents the experimental setup, including dataset details, model configuration, training settings, hardware environment, and experiment groups.

**Chapter 5** reports the experimental results across different dataset sizes, batching strategies, curriculum schedules, and generation metrics.

**Chapter 6** discusses the findings, explains why semantic batching does not consistently improve performance, and interprets the role of random batching and curriculum ordering.

**Chapter 7** concludes the thesis by summarizing the main findings, limitations, and future research directions.